

Data Flow Diagram

Date	29 September 2026
Team ID	SWTID-2026-5115
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

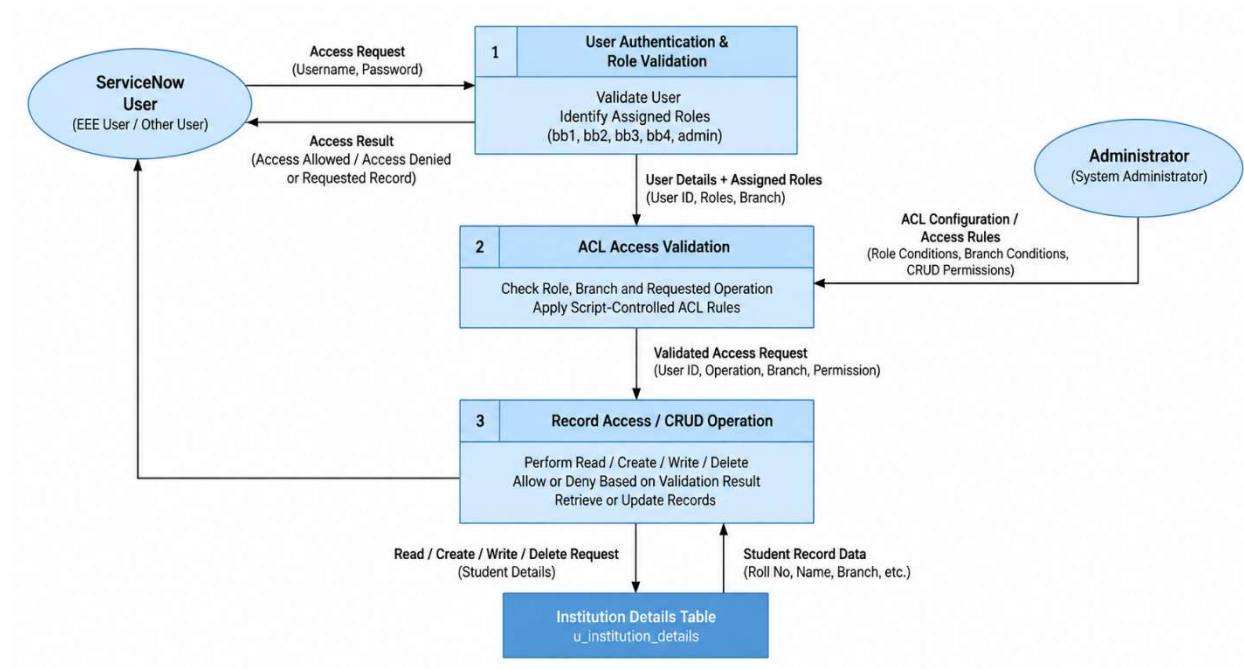
Data Flow Diagram

This data flow diagram shows how data moves when a user requests access to institution records in ServiceNow: the ServiceNow user submits an access request, the system validates the user's assigned roles and branch, and the Script-Controlled ACL determines whether the requested operation is allowed. Authorized users can perform the permitted CRUD operation on the Institution Details table, while unauthorized users are denied access. Administrators retain full access to the records.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway).
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order).
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use (e.g., Customer Database, Inventory).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

The diagram below shows how data moves between the ServiceNow user, Administrator, three access-control processes, and the Institution Details data store. The system validates user roles, branch conditions and requested operations before allowing or denying access to records.



Process Description

No.	Process	Description
1	User Authentication & Role Validation	The ServiceNow user requests access to the Institution Details records. The system identifies the user and retrieves the assigned roles such as bb1, bb2, bb3 and bb4.
2	ACL Access Validation	The Script-Controlled ACL validates the user's role, Branch value and requested operation. EEE users are allowed to access EEE records, while administrators retain full access.
3	Record Access / CRUD Operation	Based on the ACL validation result, the system allows or denies Read, Create, Write or Delete operations on the Institution Details table.